Midterm Review

In Class Exercises

Exercise 1

The recurrence $T(n) = 2T(n-1)+1$ has the solution $T(n) = O(2^n)$. Show that a substitution proof fails with the assumption $T(n) \leq c2^n$, where $c > 0$ is constant. Then show how to subtract a lower-order term to make a substitution proof work.

Exercise 2

- Sketch the recursion tree, and guess a good asymptotic bound on the solution of the recurrence below, then use the substitution method to verify your answer:

$$T(n) = T(n/2) + n^3.$$

Exercise 3

- Sketch the recursion tree, and guess a good asymptotic bound on the solution of the recurrence below, then use the substitution method to verify your answer:

$$T(n) = 4T(n/3) + n.$$

Exercise 4

- Use the master method to give tight asymptotic bounds for the following recurrences:

$$T(n) = 2T(n/4) + 1.$$

$$2T(n/4) + n^2.$$